

CITS3002: Lab 1

Exercise 1:

1-1: Check your IP configuration:

(run “ipconfig/all” for Windows or “ifconfig” for Mac/Linux in the terminal)

1-2: What is your IP address?

1-3: What is your MAC (physical) address?

1-4: Explain which layer of the OSI model each of these addresses belongs to.

Exercise 2:

Compare bus, star, and mesh topologies:

2-1: Write at least one advantage of each topology.

2-2: Write at least one disadvantage of each topology.

2-3: Which topology is most commonly used today, and why? Give an example.

Exercise 3:

For each of the protocols HTTP, TCP, and IP, identify:

3-1: Which layer does it belong to in the OSI model? What is the main functionality of each protocol?

3-2: Which layer does it belong to in the TCP/IP model?

3-3: What is the primary function of the corresponding layer in the OSI model for each protocol?